

EXPERIENCE

- **Conjecture** London, UK
Research Engineer *Aug 2023 - present*
 - Primary focus on exploratory ML experiments, including training multimodal (vision) LLMs and building distributed training infrastructure
 - Responsible for systems security and managing the k8s cluster for distributed training and inference, including working with teams to build, deploy, and scale applications
- **AI Security Bootcamp** London, UK
Program Director *June 2025 - Sept 2025*
 - Awarded a \$223,134 grant by [Open Philanthropy](#) to run the [AI Security Bootcamp](#), an in-person program for 20 participants over 4 weeks. The contents were [open sourced for self-study](#).
 - Oversaw a team of 11 staff for content development, in-person instruction, and operations
- **FlexHeg** Sheffield, UK
Consultant *Oct 2024 - Nov 2024*
 - Part of a team that received a \$200,000 grant from Davidad (ARIA program manager) to develop a flexible hardware-enabled governance (FlexHEG) prototype in 5 weeks. Built a small datacentre portable in briefcases, complete with H100s and an Infiniband network.
 - Designed a secure interface enabling AI labs and governance bodies to reach consensus on LLM training progress and implement policies. Configured ConnectX adapters to encrypt inter-GPU communication to ensure confidentiality and integrity.
- **SERI MATS** Berkeley, CA
Fellow *July 2023 - Aug 2023*
 - Mentored by Jeffrey Ladish in the [Understanding AI Hacking](#) track at Stanford Existential Risks Initiative's [ML Alignment Theory Scholars Program](#)
 - Worked on efficiently undoing RLHF from LLMs (unRLHF), resulting in [BadLLaMA](#), an unrestricted version of Meta's LLaMA 2 family of models
- **QuantCo** Zürich, Switzerland
SWE Intern *Jun 2021 - Aug 2021*
 - Developed Kubernetes toolchains to benchmark BigQuery, Microsoft SQL Server, PostgreSQL, and Snowflake databases for internal data science workloads
 - Designed QuantCo's engineering hiring challenge, challenge.quantco.com
- **KDE Community** *Nov 2018 - June 2021*
Developer; GSoC Mentor
 - **networkmanager-qt**: a Qt wrapper for NetworkManager API – during Google Code-in 2018
 - **marK**: a multimedia annotation tool for ML – during Season of KDE 2021 as a participant; and during Google Summer of Code 2021 as mentor.

FORMAL EDUCATION

- **Indian Institute of Information Technology** Lucknow, India
Bachelor of Technology in Computer Science and Artificial Intelligence; GPA: 9.2/10 *Dec. 2020 - June 2024*
Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Data Structures, Algorithms, Automata Theory, Database Management Systems, Operating Systems, Big Data Analytics, Compiler Design, Computer Graphics

PUBLICATIONS

- **BadLLaMA: Cheaply Removing Safety Fine-Tuning from Llama 2-Chat 13B** Gade P., Lermen S., Rogers-Smith C., Ladish J. arXiv preprint, 2023.
- **MORPH-EUS – Towards Folding with 1,823 Non-Canonical Amino Acids** Waldherr A., Freimann G. L., Lederbauer M., Gade P. MoML @ MIT, 2025.

SKILLS AND INTERESTS

- **Languages:** Python, TypeScript/JavaScript, Java, C++, C, Bash, Rust, Clojure, x86_64 & 8085 Assembly
- **Machine Learning:** PyTorch, CUDA, NVIDIA DOCA/DPDK, DeepSpeed, distributed training, multimodal vision LLMs, diffusion models, LLM Security, RLHF/alignment
- **Computer Engineering:** Linux, containerization, Kubernetes, Infiniband, distributed training, networking protocols, Google Cloud, git/mercurial, (La)TeX/TikZ, Android, Databases (PostgreSQL, Snowflake, MongoDB, Redis)
- Information security: **AI Security** (adversarial attacks, prompt/RAG injection, guardrail finetuning, model extraction, ablation), **reverse engineering** (Ghidra, Radare2, Android Studio, GDB), **network security** (Wireshark, Burpsuite, metasploit, aircrack, nmap, kali)

AWARDS & ACHIEVEMENTS

- **Google Code-in 2018 Grand Prize Winner** 2018
 - [Google Code-in](#) is a contest by Google to encourage participation in open-source software. Completed 34 tasks in the contest and was selected as the Grand Prize winner by [KDE Community](#).
 - As part of the prize, invited to Google's Headquarters in Mountain View, California.
- **Ranked 52 out of 15817 globally in PicoCTF** 2019
 - CTFs are cybersecurity competitions consisting of a series of challenges where participants must reverse engineer, break, hack, or decrypt the challenges. PicoCTF is organized by security experts at Carnegie Mellon University.
 - Team ranked 52nd out of 15817 among high school and college students from around the world.

PROJECTS

- **microcommit.io:** Microcommit enhances organizational coordination with 5-minute micro-requests for volunteers.
- **clippy:** Clippy is a browser extension that clips content you teach it to hide from you.
- **ghsok:** Gin Hyperparameter Search On Kubernetes automates jobs on k8s clusters to search over hyperparameters
- **glide:** Implementation of [OpenAI's GLIDE](#) text-to-image synthesis algorithm
- **glide-train:** Training code implementation for [GLIDE](#)
- **jLHS:** Fast and tiny HTTP/1.1 server written in Java without any dependencies, supports GET and POST requests
- **Java-Disassembler:** .class file parser and serializer. Also disassembles code from compiled Java code.
- **FabricHax:** A Minecraft mod to automate repetitive tasks
- **PhonePut:** An Android application that turns a device into an input device for a computer
- **htccpcp-impl:** Implementation of RFC 2324 – Hyper Text Coffee Pot Control Protocol (HTCPCP/1.0)
- **networkmanager-qt:** NetworkManagerQt provides access to all NetworkManager features exposed to DBus API
- **marK:** A machine learning dataset annotation tool which allows users to annotate multiple types of data
- **pranavg.me:** Personal website